KALAIGNARKARUNANIDHI INSTITUTE OF TECHNOLOGY, COIMBATORE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BLOCKCHAIN-BASED SECURE VOTING SYSTEM

"Your Vote. Your Voice. Securely."

A Database Management Systems (DBMS) Mini-Project Report
submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING (B.E.) IN COMPUTER SCIENCE AND ENGINEERING

Submitted By:

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Class: CSE-C | Academic Year: 2025–2029

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BONAFIDE CERTIFICATE

This is to certify that the project report titled **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"** is a bonafide record of independent project work carried out by **MEGA SREE MS** (Register Number: **711525BCS092**), student of the Department of Computer Science and Engineering, Kalaignarkarunanidhi Institute of Technology, Coimbatore, during the academic year 2025–2029, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering (B.E.) in Computer Science and Engineering**.

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[Project Guide Name] Project Guide & Assistant Professor Department of CSE	**[Head of the Department]** Professor & Head of Department Department of CSE
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STUDENT DECLARATION

I, **MEGA SREE MS** (Register Number: **711525BCS092**), hereby declare that the project entitled **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"** submitted to **Kalaignarkarunanidhi Institute of Technology, Coimbatore** in partial fulfillment of the requirements for the award of **Bachelor of Engineering (B.E.) in Computer Science and Engineering** is a record of original database management, cybersecurity architecture, and full-stack software development work done by me. I confirm that this report has not been submitted previously for any degree, diploma, or academic award at any other educational institution or university.

Place: Coimbatore
Date: October 2026

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ABSTRACT

Electronic voting platforms represent the vanguard of modern democratic engagement but frequently suffer from critical vulnerabilities including duplicate ballot submission, centralized administrative database tampering, and voter privacy compromise. This academic project presents the design and implementation of the **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"**, a secure, full-stack, and user-centric platform tailored for collegiate and organizational governance under the tagline *"Your Vote. Your Voice. Securely."*.

The architecture bridges a normalized relational Database Management System (3NF schema using SQLite3 / MySQL) with a custom SHA-256 hash-linked cryptographic ledger. Voter eligibility and identity verification are mathematically decoupled from cast ballots, ensuring complete ballot anonymity while strictly enforcing the fundamental "one vote per eligible voter" constraint through database-level transactions. Each accepted ballot is mined into an append-only block containing the preceding block's cryptographic hash, enabling real-time verification and immediate detection of unauthorized data modifications.

Furthermore, the project delivers an advanced user experience featuring a high-contrast White Mode interface by default, instant Dark Mode toggling, and an integrated AI Voice Assistant powered by the Web Speech API capable of voice navigation, candidate manifesto readout, and real-time ledger verification audits.

Keywords: Relational DBMS, 3NF Normalization, Blockchain Ledger, SHA-256 Hash Linking, Anonymous Ballot, AI Voice Assistant, White Mode UI, ACID Transactions, Antigravity.

CHAPTER 1: INTRODUCTION

1.1 Background of Electronic Voting

Democratic governance relies fundamentally on the integrity, accessibility, and transparency of election mechanisms. Over the past three decades, electoral bodies globally have transitioned from manual paper ballots to electronic voting systems. Electronic voting encompasses Direct Recording Electronic (DRE) systems, optical scan tabulation, and remote internet-based balloting portals. While electronic mechanisms dramatically accelerate tallying and reduce paper wastage, they introduce grave cybersecurity vulnerabilities, including single points of failure, unauthorized database alterations, and vulnerabilities to insider coercion.

1.2 Problems with Conventional Voting Systems

Conventional electronic voting solutions exhibit significant systemic drawbacks: (1) Vulnerability to insider manipulation where a database administrator can alter records; (2) Absence of ballot secrecy guarantees when voter IDs are stored directly beside ballot selections; (3) Susceptibility to race conditions and duplicate voting when transactions are non-atomic; and (4) Lack of verifiable mathematical auditability.

1.3 Objectives and Scope

Developed by **MEGA SREE MS** (711525BCS092) at **Kalaignarkarunanidhi Institute of Technology, Coimbatore**, the project objectives include: implementing a 3NF normalized relational schema, enforcing atomic ACID voting transactions, decoupling voter identity from cast ballots, building an append-only SHA-256 cryptographic ledger with genesis block initialization, delivering a radiant White Mode interface with Dark Mode toggling, and integrating a Web Speech API AI Voice Assistant for multimodal accessibility.

CHAPTER 2: LITERATURE REVIEW

2.1 Conventional vs. Blockchain-Based Voting

Studies by Kohno et al. and Springall et al. demonstrated that centralized election databases are vulnerable to internal database manipulation. Blockchain technology structures data into an append-only chain of blocks where block i references hash(block i - 1). Using SHA-256 (NIST FIPS 180-4), modifying any stored ballot alters its digest completely, creating an immediate pointer mismatch with all following blocks.

2.2 Summary of Technical Differences

Table 2.1 highlights the comparative advantages of the proposed hybrid relational-blockchain architecture over legacy systems.

CHAPTER 3: SYSTEM REQUIREMENTS AND DESIGN

3.1 3NF Normalization Analysis

The database schema is normalized up to Third Normal Form (3NF):

• **1NF:** Atomic columns, unique surrogate integer primary keys across all seven tables.

• **2NF:** Full functional dependency on primary keys; associative entities use compound UNIQUE constraints.

• **3NF:** Elimination of transitive dependencies; candidate details, election details, and voter profiles reside in distinct entities.

3.2 Anonymous Ballot Model

When a vote is cast, `voter_eligibility.has_voted` is set to 1 for the voter, while the ballot choice is stored in `votes` with only `candidate_id` and an immutable cryptographic `block_id`. The voter's identity is never linked to the ballot.

CHAPTER 4: BLOCKCHAIN & CRYPTOGRAPHIC LEDGER

4.1 SHA-256 Block Structure

Each block record contains: block_index, previous_hash, current_hash, transaction_hash, and created_at. Genesis block (index 0) uses 64 zeros as its previous hash.

4.2 Verification and Tamper Detection

The verification endpoint recalculates every block's hash and checks sequential pointer continuity: block[i].previous_hash == block[i-1].current_hash. Any retroactive database tampering flags the exact broken block.

CHAPTER 5: TESTING AND EVALUATION

5.1 Automated Test Suite Execution Results

The test suite executed via npm test validates 4 critical assertions:

• **TC-01 (Hash Determinism):** SHA-256 produces identical 64-char hex strings for identical block inputs. (PASS)

• **TC-02 (Clean Ledger Verification):** Sequential block index and previous-hash pointers validate 100% intact. (PASS)

• **TC-03 (Duplicate-Vote Prevention):** Second ballot attempt by same voter triggers database constraint rollback. (PASS)

• **TC-04 (Tamper Detection):** Altering previous hash in block #1 flags exact corrupted block. (PASS)

5.2 Summary of Genuine Test Metrics

All 4 automated tests passed in 80.6ms. The system successfully prevented duplicate votes and detected simulated ledger tampering.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Summary of Contributions

The **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"** developed by **MEGA SREE MS** (Register Number: **711525BCS092**) at **Kalaignarkarunanidhi Institute of Technology, Coimbatore** successfully proves how relational DBMS normalization, ACID transactions, and SHA-256 hash chaining converge to safeguard electronic democratic voting.

6.2 Future Enhancements

Future enhancements include integrating Zero-Knowledge Proofs (zk-SNARKs) to provide cryptographically blind receipts to voters and expanding the backend to multi-node Byzantine Fault Tolerant (BFT) consensus networks.